Matrix Mult. General Rule : $\rightarrow C_{ij} = A_{i1}B_{1j} + A_{i2}B_{2j} + \dots + A_{ip}B_{pj}$] Row i of A, Column j of B] → dot product

Hw9 con.

C31. $A = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$ $\begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}$

$A^2 = A \cdot A$

$$A^2 = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1\cdot 1 + (-1)\cdot 0 & 1\cdot(-1) + (-1)\cdot 1 \\ 0\cdot 1 + 1\cdot 0 & 0\cdot(-1) + 1\cdot 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$$

$$A^3 = A^2 \cdot A = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1\cdot 1 + (-2)\cdot 0 & 1\cdot(-1) + (-2)\cdot 1 \\ 0\cdot 1 + 1\cdot 0 & 0\cdot(-1) + 1\cdot 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -3 \\ 0 & 1 \end{bmatrix}$$

$$A^4 = A^3 \cdot A = \begin{bmatrix} 1 & -3 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1\cdot 1 + (-3)\cdot 0 & 1\cdot(-1) + (-3)\cdot 1 \\ 0\cdot 1 + 1\cdot 0 & 0\cdot(-1) + 1\cdot 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -4 \\ 0 & 1 \end{bmatrix}$$

Pattern: $A^n = \begin{bmatrix} 1 & -n \\ 0 & 1 \end{bmatrix}$ $A^{n+1} = \begin{bmatrix} 1 & -(n+1) \\ 0 & 1 \end{bmatrix}$

- Diagonal remains at one
- Top right remains at a $-n$
- Bottom left # remains at zero
- $A = I + N$ with $N = \begin{bmatrix} 0 & -1 \\ 0 & 0 \end{bmatrix}$. $N^2 = 0$

  $A^n = (I + N)^n = I + nN$

Higher powers of n vanish